

MTH:3340 Abstract Algebra Quiz 4

Due: Friday Feb. 27

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: Your ability to express your solutions is just as important as mathematical accuracy. You are expected to write every solution using complete sentences when possible. You will be penalized up to 4 points for notation and disorganized or hard to read solutions.

1. (10 points) Let $(G, \circ)$ be a group with identity e . Show that the subset of G defined as $Z(G) = \{x \in G \mid g \circ x = x \circ g \text{ where } g \in G\}$ is a subgroup of G .

* $Z(G) \subseteq G$ by its definition.

Since $e \in G$, for any other $g \in G$ $g \circ e = e \circ g$

so $e \in Z(G)$, thus $Z(G)$ is non-empty.

Subgroup Test: Let $x, y \in Z(G)$ and g be any element of G . Then

$$g \circ (x \circ y^{-1}) = (y^{-1} \circ y) \circ g \circ (x \circ y^{-1}) = y^{-1} \circ (y \circ g) \circ (x \circ y^{-1})$$

$$\stackrel{*}{=} y^{-1} \circ (g \circ y) \circ (x \circ y^{-1}) = (y^{-1} \circ g) \circ (y \circ x) \circ y^{-1}$$

since $y \in Z(G)$ and $g \in G$

$$\stackrel{*}{=} (y^{-1} \circ g) \circ (x \circ y) \circ y^{-1} = y^{-1} \circ (g \circ x) \circ (y \circ y^{-1})$$

since $x \in Z(G)$ and $y \in G$

$$\stackrel{*}{=} y^{-1} \circ (x \circ g) = (y^{-1} \circ x) \circ g \stackrel{*}{=} (x \circ y^{-1}) \circ g$$

since $x \in Z(G)$ and $g \in G$

since $x \in Z(G)$ and $y^{-1} \in G$

Thus $x \circ y^{-1} \in Z(G)$ so $Z(G) \leq G$.

2. (16 points) Let $(G, \circ)$ be a group and let H be any subgroup of G . For a fixed element g in G , define the set $K = \{k \mid k = g \circ h \circ g^{-1} \text{ for all } h \in H\}$. The set K is often denoted by $K = gHg^{-1}$.

(a) Show that K is also a subgroup of G .

* Since $H \leq G$, the identity $e \in G$ is also in H . Then $g \circ e \circ g^{-1} = e \in K$ so K is non-empty and K is a subset of G by definition.

Subgroup Test: Let $k_1, k_2 \in K$. Then there are $h_1, h_2 \in H$ so that $k_1 = g \circ h_1 \circ g^{-1}$ and $k_2 = g \circ h_2 \circ g^{-1}$.

$$\begin{aligned} k_1 \circ k_2^{-1} &= (g \circ h_1 \circ g^{-1}) \circ (g \circ h_2 \circ g^{-1})^{-1} = (g \circ h_1 \circ g^{-1}) \circ ((g^{-1})^{-1} \circ h_2^{-1} \circ g^{-1}) \\ &= g \circ h_1 \circ (g^{-1} \circ g) \circ h_2^{-1} \circ g^{-1} = g \circ (h_1 \circ h_2^{-1}) \circ g^{-1}. \end{aligned}$$

Since $H \leq G$, $h_1 \circ h_2^{-1} = h_3 \in H$ so

$$k_1 \circ k_2^{-1} = g \circ h_3 \circ g^{-1} \in K.$$

Therefore $K \leq G$.

- (b) Let $G = GL_2(\mathbb{R})$ and H be the subgroup of all invertible upper-triangular matrices,
 $H = \left\{ \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \mid \text{where } a, b, c \in \mathbb{R} \text{ and } ac \neq 0 \right\}$. If we fix $g = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \in G$, describe the elements
of $K = gHg^{-1}$ in words and in set-builder notation.

If h is any element of H . Then

$$h = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \text{ for some } a, b, c \in \mathbb{R} \text{ where } ac \neq 0.$$

$$\text{Since } \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

$$\begin{aligned} \text{Then } g \circ h \circ g^{-1} &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & c \\ a & b \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} c & 0 \\ b & a \end{bmatrix} \end{aligned}$$

So K is the set of all ^{*}invertible lower-triangular matrices with real entries.
 $\downarrow$
since $\det \left(\begin{bmatrix} c & 0 \\ b & a \end{bmatrix} \right) = c \cdot a \neq 0$.

$$K = \left\{ \begin{bmatrix} a' & 0 \\ b' & c' \end{bmatrix} \mid a', b', c' \in \mathbb{R} \text{ and } a'c' \neq 0 \right\}$$